Review:

Markov chain

① Defn: Markovian Property: Conditional Indepent.

② Specification: S: State space.

T: Index set

$\mathbb{P}^{n,n+1}$ one-step transition prob. matrices

$\mathbb{P}^{n,n+1} = \mathbb{P}$, stationary MC.

③ Short term performance

a. $P(X_3=j \mid X_0=i)$?

b. $P(X_3=j \mid X_0 \sim \pi_0)$?

$\Rightarrow$ dist. of X_3? expectation? Var? pgf?

c. $P(X_0=i, X_1=j, X_2=k)$?

$P(X_1=i, X_2=j, X_3=k)$?

d. $P(X_1=i, X_3=k, X_7=j)$?

path

e. $P(X_1=i \mid X_3=k, X_0=j)$?

④ First Step Analysis

$$a_i = \sum P_{ij} a_j + g(i)$$

⑤ Long-run performance

Irreducible MC

$|S| = \infty$

$m_j = \infty$, $\pi_j = 0$

$m_j < \infty$ positive recur.

transient $\pi_j = 0$

$|S| < \infty$

period $=1$: $\pi = \pi P$, $\sum \pi_i = 1$

period >1: no limiting. but stationary dist $\pi = \pi P$

m_j: mean duration between two visits of state j.

$$\frac{\sum P_{ij}^{(n)}}{n} \xrightarrow{n\to\infty} \frac{1}{m_j}$$

Reducible MC.

→ Classes → Recurrent ($f_{ii} \to N_i = \infty$) $\xrightarrow{\text{each class}}$ irreducible MC. (last page) $\Rightarrow \pi_\infty | \pi_2$

→ Transient ($f_{ii} \to N_i = \frac{f_{ii}}{1-f_{ii}}$) $\xrightarrow{\text{each state}}$ ~~eh~~ P(entering each recur. class)

Irreducible

$$\lim_{n\to\infty} P_{ij}^{(n)} = \pi_j$$

$$P^{(n)} \xrightarrow{n\to\infty} \begin{pmatrix} \pi_1 & \pi_2 & \cdots & \pi_{|S|} \\ \pi_1 & \pi_2 & & \vdots \\ \vdots & \vdots & & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_{|S|} \end{pmatrix}$$

Reducible

$$\lim_{n\to\infty} P_{ij}^{(n)} = \pi_{j|i}$$

$$P^{(n)} \xrightarrow{n\to\infty} \begin{pmatrix} \pi_{1|1} & \cdots & \\ \pi_{2|1} & \cdots & \\ \vdots & \ddots & \\ \pi_{|S|1} & \cdots & \end{pmatrix}$$

$i \Leftrightarrow j$ recur. class

AFI.

Applications.

BP → P.g.f. and properties

→ $E[X_n]$ $Var(X_n)$ depends on $E[\xi]$, $Var(\xi)$

→ $f_{X_n}(t)$ $f_\xi(t)$

→ Extinction prob. $x = f_\xi(x)$

~~Ro~~ Probabilistic Structure: $X_0 = k$ indep k BP with ~~k=~~ $X_0 = 1$

PageRank: Network of websites

MC: $\{S, T, P\}$?

limiting dist. → π_∞ → Ranking

MCMC Sampling $\pi = \pi P$ Global Balanced Equations

$\pi_i P_{ij} = \pi_j P_{ji}$ Local - - - - - - - -

Q, α, ⇒ $\{S, T, P\}$ MC.

??

$X_1 \cdots X_{500}$ → Truncation step

→ gap.

Poisson Process

① Defn. ① ② ③

② Properties a. $X(t) \sim Pois(\lambda t)$

$P(X(t) = k)$

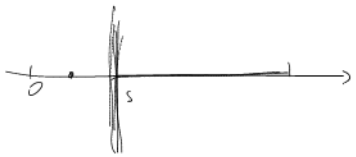
b. Indep between disjoint time intervals

$P(X(s) = k, X(t) = m)$

$P(X(s) = k \mid X(t) = m)$ for $s < t$.

c. Waiting time.

$W_1 \sim Exp(\lambda)$ $S_i \sim Exp(\lambda)$ iid

d. $W_1 \mid X(t) = 1 \sim Unif(0, t)$

e.